

This phase focuses on expanding the ULTIMA 2.0 operating system simulator through the implementation of a Memory Management Unit or MMU. The system introduces a structured memory allocation model where each task can request, use, read, and write, and release memory through a controlled interface. Memory is managed within a fixed-space and is divided into blocks that are assigned to tasks using a first-fit allocation strategy. Each allocation returns a unique memory handle that is stored within the task's control block, allowing the system to track ownership and usage of memory throughout execution.

To support this functionality, a memory management system was developed that provides allocation and deallocation operations, as well as single and multi-byte read and write capabilities. The MMU maintains a linked list of memory segments to track free and used regions, along with detailed information such as starting location, size, and current position within each block. Additional debugging features such as memory dumps and memory usage tables allow the system to display the current state of memory, making it easier to visualize allocation, deallocation, and coalescing of memory blocks.

The MMU is integrated into the scheduler environment established in Phase 1 and works alongside the IPC system from Phase 2, allowing tasks to manage memory while executing and communication concurrently. Access to shared memory structures is protected using a Core Memory semaphore, ensuring safe synchronization and preventing race conditions during allocation and freeing operations. This phase also introduces basic memory protection by restricting tasks to only access memory associated with their own task ID and memory handle, simulating base and limit register behavior.